

Nikhil D'Souza

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Education

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- Columbia University** – New York City, NY Dec 2023
- M.S. in Data Science
 - Coursework: Probability & Statistics, Applied Machine Learning, Mathematics of Finance, Finance & Structuring for Data Science
- Purdue University** – West Lafayette, IN May 2022
- B.S. in Computer Science, Data Science, Applied Statistics (GPA: 3.84)
 - Coursework: Advanced Neural Projects, Data Mining & ML, Analysis of Algorithms, Neural Image Processing, Intro to AI, Data Structures & Algorithms, Large-Scale Data Analytics, Regression Analysis, Probability, Linear Algebra, Multivariable Calculus
 - Head Teaching Assistant of The Data Mine learning community (1000+ students and 50+ corporate partners)
- Coursera** – DeepLearning.AI by Andrew Ng Aug 2020
- Neural Networks and Deep Learning
 - Improving Deep Neural Networks: Hyperparameter Tuning, Regularization, and Optimization

Experience

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- Vitalize (YC W23)** – Cofounder & CTO Feb 2022 - Present
- Built the first digital wellness platform for the healthcare workforce (vitalizecare.co).
 - Developed a clinician-centered mobile app and web-based admin dashboard using MongoDB, Express.js, React, and Node.js.
 - Led development team to build anonymous video coaching, dynamic surveys, and analytics dashboard.
- Fast MRI Reconstruction using Deep Learning** – Undergraduate Researcher Aug 2021 - May 2022
- Investigated the use of AI to make MRI scans up to 10x faster by reconstructing accurate images from undersampled data.
 - Trained a generative adversarial network with images that are Fourier transforms of 20% subsampled and fully sampled k -space data.
 - Utilized a GAN with a U-Net-based generator architecture and residual-in-residual dense blocks (RRDBs).
- Eli Lilly** – AI/ML Intern May - Aug 2021
- Analyzed clinical trial data to predict trial site performance and identify new indications of in-market Lilly drugs.
 - Used PCA, logistic regression, and XGBoost to determine if Lilly-sponsored trials should continue being funded.
 - Identified reasons why investigators reforecast trial enrollment estimates using TF-IDF.
- Vincere Health** – Software Engineer & Data Scientist Oct 2019 - Aug 2020
- Built a platform that uses financial incentives, behavioral nudges, and evidence-based interventions to help quit smoking and be healthy.
 - Developed patient facial recognition, chat messaging using sockets, and incentive engine using Redis.
 - Spearheaded data science capability providing actionable insights, user performance, and patient rewards.
- Saama** – AI Intern Jun - Aug 2018
- Used supervised ML techniques to identify packaging anomalies on a conveyor belt for a leading biopharmaceutical company.
 - Developed CNN with 99% validation accuracy and leveraged image augmentation to expand the dataset.

Projects and Awards

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- Mask Autoencoder** – Smart Software Hackathon (2nd Place Overall Winner) 2021
- Feverbase** – COVID clinical trial data platform 2020
- Harvard President's Innovation Challenge** – Grand Prize Winner 2020
- L3Harris Technologies Scholarship** – Purdue CS Awards 2019
- Calculating Fatigue of Rugby Athletes** – ASA DataFest (1st Place Overall Winner) 2019
- Predicting Emerging eSport Celebrities** – Krannert eSport Hackathon (3rd Place Overall Winner) 2018
- GluClose** – PennApps XVIII (2nd Place Overall Winner) 2018
- Apple Worldwide Developer Conference Scholarship Winner** 2015

Skills and Technologies

Languages/Tools: Python, R, SQL, Tensorflow, Pytorch, Flask, React, Typescript, JS ES6, Node.js, Swift, Java, C, Unix, Git, Docker, Redis

AWS: EC2, S3, Lambda, API Gateway, Sagemaker, Elasticsearch, CloudWatch, Rekognition, Comprehend, SNS, Cognito